北京大學
PEKING UNIVERSITY

第二章 原子结构与光谱

1. 类氢原子

原子模型的演变

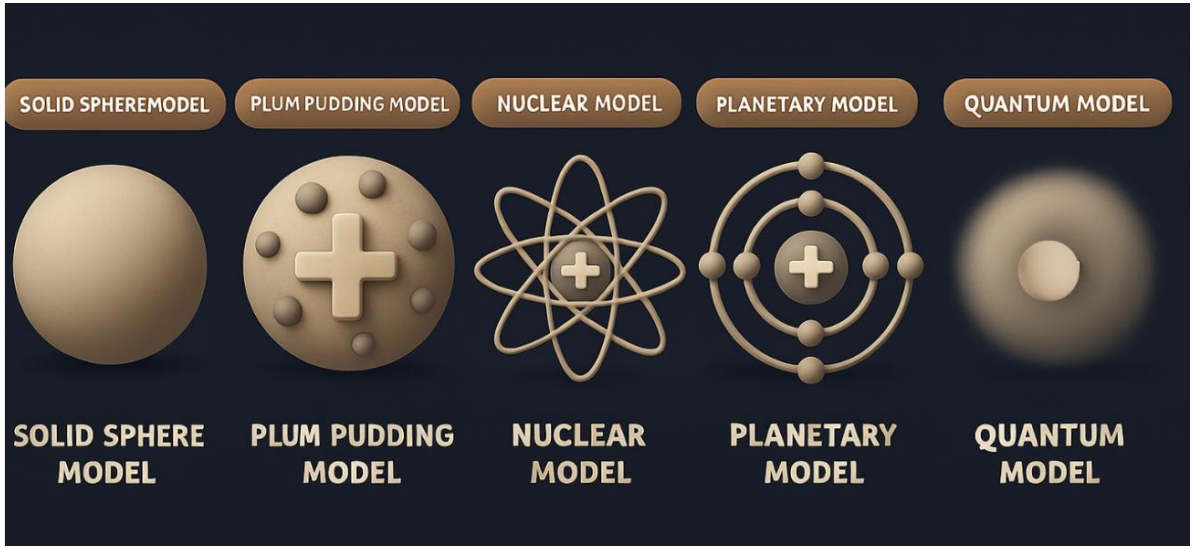
Dalton（1803）

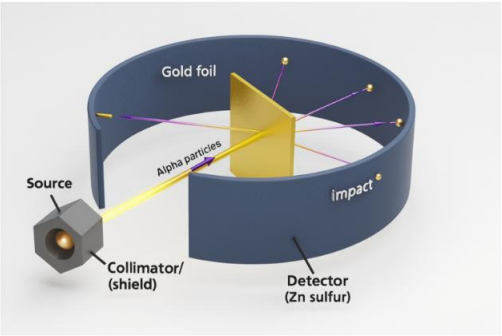
J. J. Thomson（1897）

Rutherford (1911)

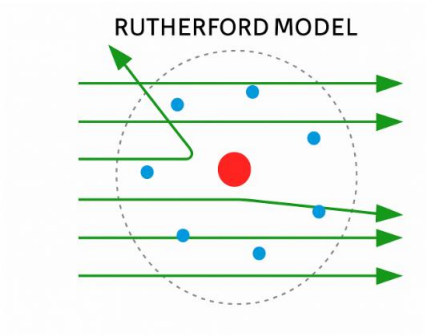
Bohr (1913)

Schrödinger (1925)

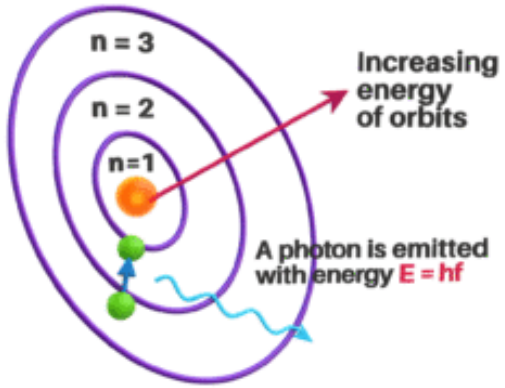


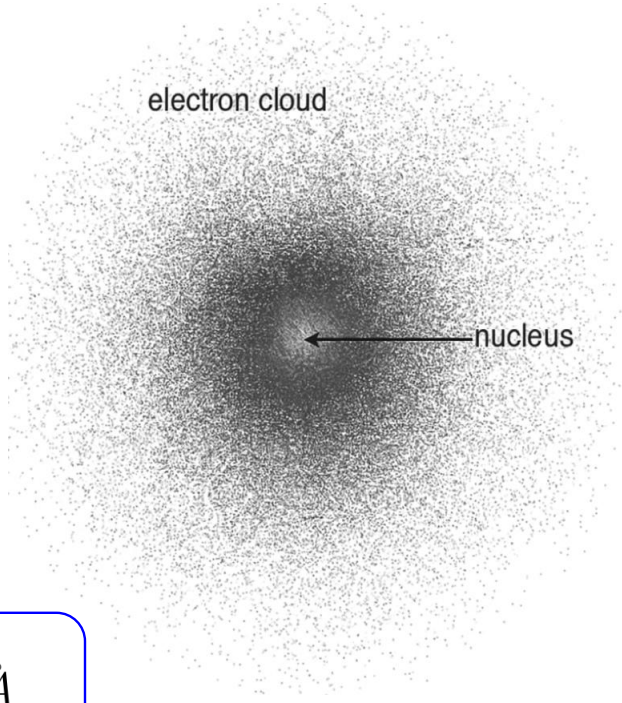


gold foil experiment



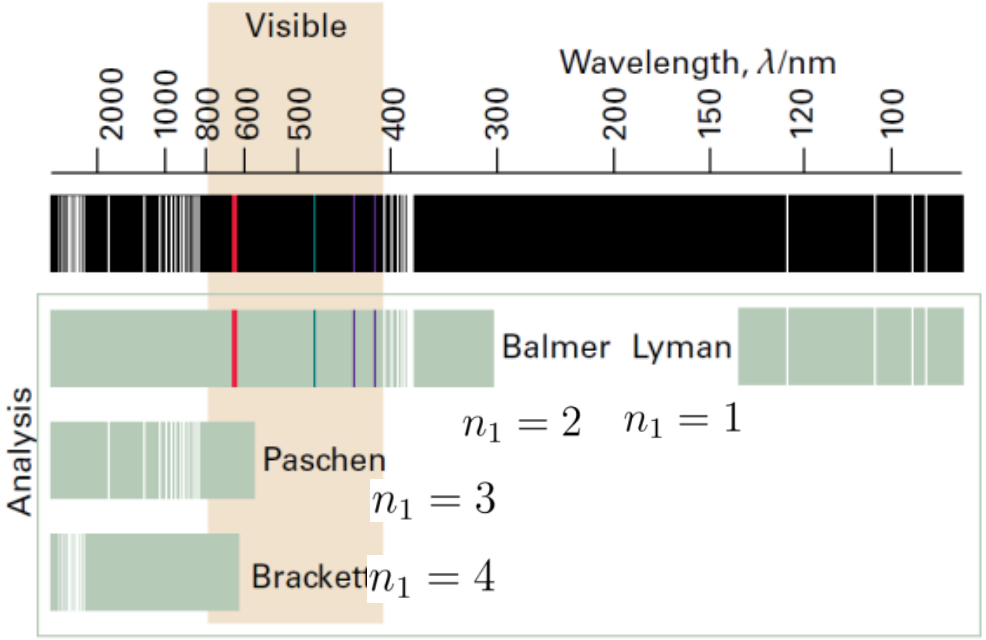
Rutherford Model





氢原子光谱线规律 $\lambda = B\frac{n^2}{n^2-4}$ $B = 3646\mathring{A}$

类氢原子：单电子原子，核电荷Z



电子束轰击H_2气体，导致氢分子解离

获得**激发态**的H原子，发射电磁波，观测得到氢原子光谱

波数满足里德堡(Rydberg)关系

$$\tilde{\nu} = \tilde{R}_H \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

$$\tilde{R}_H = 109677\mathrm{cm}^{-1}$$ 里德堡常数

氢原子哈密顿量

$$\hat{H} = \hat{E}_{\mathrm{k,electron}} + \hat{E}_{\mathrm{k,nucleus}} + \hat{V}(r)$$

$$= -\frac{\hbar^2}{2m_e}\nabla_e^2 - \frac{\hbar^2}{2m_N}\nabla_N^2 - \frac{Ze^2}{4\pi\varepsilon_0 r}$$

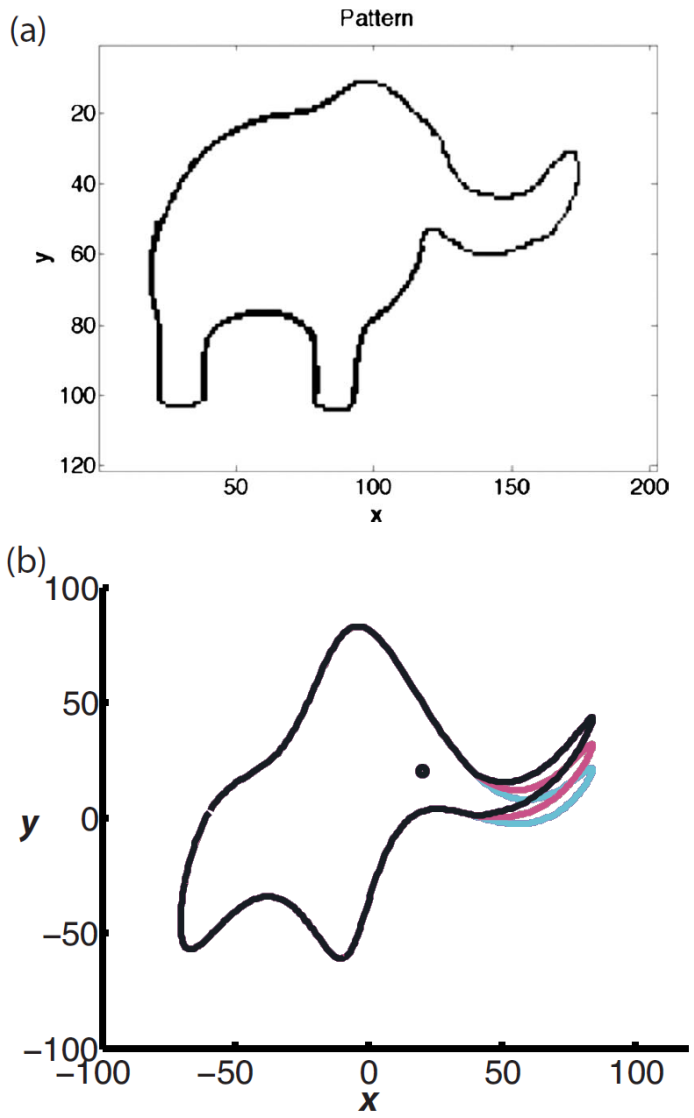
原子核与电子都是变量

With four parameters I can fit an elephant, and with five I can make him wiggle his trunk.

-- von Neumann

There are two ways of doing calculations in theoretical physics. One way, and this is the way I prefer, is to have a clear physical picture of the process you are calculating. The other way is to have a precise and self-consistent mathematical formalism.

-- Enrico Fermi



F. Dyson, "A meeting with Enrico Fermi," Nature, 427(6972), 297, 2004.

J. Mayer, K. Khairy, K. Howard, Am. J. Phys. 78, 6, 2010.

质心坐标与相对位移

$$\boldsymbol{R} = \frac{m_e \boldsymbol{r}_e + m_N \boldsymbol{r}_N}{m_e + m_N}$$

$$\boldsymbol{r} = \boldsymbol{r}_e - \boldsymbol{r}_N$$

改写动能项 = 质心动能 + 相对动能

$$-\frac{1}{2m_e}\nabla_e^2 - \frac{1}{2m_N}\nabla_N^2 = -\frac{1}{2M}\nabla_{\boldsymbol{R}}^2 - \frac{1}{2\mu}\nabla_{\boldsymbol{r}}^2$$

$$M = m_e + m_N \qquad \mu = \frac{m_e m_N}{m_e + m_N}$$

$$-\frac{\hbar^2}{2M}\nabla_{\boldsymbol{R}}^2\Psi - \frac{\hbar^2}{2\mu}\nabla_{\boldsymbol{r}}^2\Psi - \frac{Ze^2}{4\pi\epsilon_0 r}\Psi = E_{\text{tot}}\Psi$$

变量分离，令

$$\Psi(\boldsymbol{r}, \boldsymbol{R}) = \chi(\boldsymbol{R})\psi(\boldsymbol{r})$$

$$E_{\text{tot}} = \mathcal{E} + E$$

$$-\frac{\hbar^2}{2M}\nabla_{\boldsymbol{R}}^2\chi = \mathcal{E}\chi$$

$$-\frac{\hbar^2}{2\mu}\nabla_{\boldsymbol{r}}^2\psi - \frac{Ze^2}{4\pi\epsilon_0 r}\psi = E\psi$$

质心波函数满足自由粒子运动方程，解为

$$\chi(\boldsymbol{R}) = \frac{1}{(2\pi)^{3/2}}e^{i\boldsymbol{k}\cdot\boldsymbol{R}} \qquad \mathcal{E} = \frac{\hbar^2 k^2}{2M}$$

（一维自由粒子推广到三维）

相对运动方程中 $\mu \approx m_e$

利用球坐标，可进一步分离变量

$$\nabla^2 = \frac{1}{r}\frac{\partial^2}{\partial r^2}r + \frac{1}{r^2}\Lambda^2$$

$$\Lambda^2 = \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2} + \frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\sin\theta\frac{\partial}{\partial\theta}$$

$$\psi(\boldsymbol{r}) = \psi(r, \theta, \phi) = R(r)Y(\theta, \phi)$$

$$-\frac{\hbar^2}{2\mu}\nabla_{\boldsymbol{r}}^2\psi-\frac{Ze^2}{4\pi\epsilon_0 r}\psi=E\psi$$

$$-\frac{\hbar^2}{2\mu}\left[\frac{1}{r}\left(\frac{\partial^2}{\partial r^2}r\right)+\frac{\hat{\Lambda}^2}{r^2}\right]R(r)Y(\theta,\phi)-\frac{Ze^2}{4\pi\epsilon_0 r}R(r)Y(\theta,\phi)=ER(r)Y(\theta,\phi)$$

等式两边同除以ψ，得

$$-\frac{\hbar^2}{2\mu}\frac{1}{rR(r)}\left(\frac{\partial^2}{\partial r^2}rR(r)\right)-\frac{\hbar^2}{2\mu}\frac{1}{r^2Y(\theta,\phi)}\hat{\Lambda}^2Y(\theta,\phi)-\frac{Ze^2}{4\pi\epsilon_0 r}=E$$

等式两边同乘以r^2，整理得

$$-\frac{\hbar^2}{2\mu}\frac{r}{R(r)}\left(\frac{\partial^2}{\partial r^2}rR(r)\right)-\frac{Ze^2r}{4\pi\epsilon_0}-Er^2=\frac{\hbar^2}{2\mu}\frac{1}{Y(\theta,\phi)}\hat{\Lambda}^2Y(\theta,\phi)=C\frac{\hbar^2}{2\mu}$$

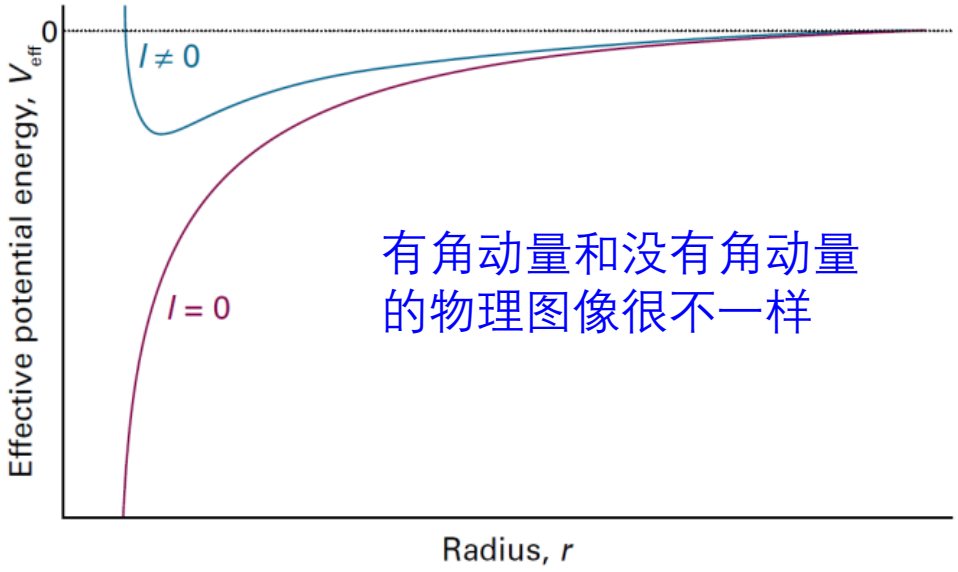
$$\Longrightarrow\quad \hat{\Lambda}^2Y(\theta,\phi)=CY(\theta,\phi)\qquad\Longrightarrow\quad Y(\theta,\phi)=Y_{l,m_l}(\theta,\phi)\quad C=-l(l+1)$$

代回整理得径向方程

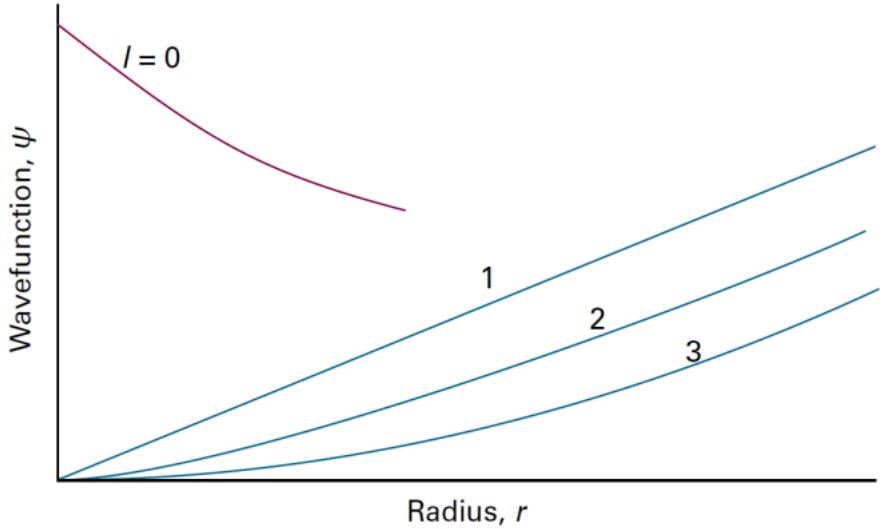
$$-\frac{\hbar^2}{2\mu}\left(\frac{\partial^2R}{\partial r^2}+\frac{2}{r}\frac{\partial R}{\partial r}\right)+\left(\frac{l(l+1)\hbar^2}{2\mu r^2}-\frac{Ze^2}{4\pi\epsilon_0 r}\right)R(r)=ER(r)$$

径向方程 $$-\frac{\hbar^2}{2\mu}\left(\frac{\mathrm{d}^2R}{\mathrm{d}r^2}+\frac{2}{r}\frac{\mathrm{d}R}{\mathrm{d}r}\right)+V_{\mathrm{eff}}R=ER$$

$$V_{\mathrm{eff}}(r)=-\frac{Ze^2}{4\pi\varepsilon_0 r}+\frac{l(l+1)\hbar^2}{2\mu r^2}$$



有角动量和没有角动量的物理图像很不一样



- $r \to \infty$，主要感受到库伦势尾巴，波函数指数衰减，速度与l无关

$$R(r)=\overbrace{r^l}^{\text{Dominant close to the nucleus}}\times\overbrace{(\text{polynomial in } r)}^{\text{Bridges the two ends of the function}}\times\overbrace{(\text{decaying exponential in } r)}^{\text{Dominant far from the nucleus}}$$

- $r \to 0$，$l \neq 0$时势能趋于无穷，波函数趋于0

  l 越大，波函数趋于0速度越快

$$R(r) = r^l L(r) e^{-\alpha r}$$

$$-\frac{\hbar^2}{2\mu}\left(\frac{\partial^2 R}{\partial r^2}+\frac{2}{r}\frac{\partial R}{\partial r}\right)+\left(\frac{l(l+1)\hbar^2}{2\mu r^2}-\frac{Ze^2}{4\pi\epsilon_0 r}\right)R(r)=ER(r)$$

为简化方程，作如下变元

$$\rho=\frac{2Zr}{na} \qquad a=\frac{m_e}{\mu}a_0 \qquad a_0=\frac{4\pi\varepsilon_0\hbar^2}{m_e e^2}$$

$$R(r)=r^l L(r)e^{-\alpha r}$$

本征波函数满足

$$R_{n,l}(r)=N_{n,l}\rho^l L_{n,l}(\rho)e^{-\rho/2}$$

$L_{n,l}$是连带拉盖尔多项式 $x\frac{d^2L}{dx^2}+(l+1-x)\frac{dL}{dx}+nL=0$

n必须为整数——量子化条件

$$E_n=-\frac{\mu e^4}{32\pi^2\varepsilon_0^2\hbar^2}\times\frac{Z^2}{n^2}$$ 束缚态能量

n	l	$R_{n,l}(r)$
1	0	$2\left(\frac{Z}{a}\right)^{3/2}e^{-\rho/2}$
2	0	$\frac{1}{8^{1/2}}\left(\frac{Z}{a}\right)^{3/2}(2-\rho)e^{-\rho/2}$
2	1	$\frac{1}{24^{1/2}}\left(\frac{Z}{a}\right)^{3/2}\rho e^{-\rho/2}$
3	0	$\frac{1}{243^{1/2}}\left(\frac{Z}{a}\right)^{3/2}(6-6\rho+\rho^2)e^{-\rho/2}$
3	1	$\frac{1}{486^{1/2}}\left(\frac{Z}{a}\right)^{3/2}(4-\rho)\rho e^{-\rho/2}$
3	2	$\frac{1}{2430^{1/2}}\left(\frac{Z}{a}\right)^{3/2}\rho^2 e^{-\rho/2}$

$$\int_0^\infty R_{n,l}(r)^2 r^2 dr=1$$

径向波函数归一条件

注意：$n\geq l+1$

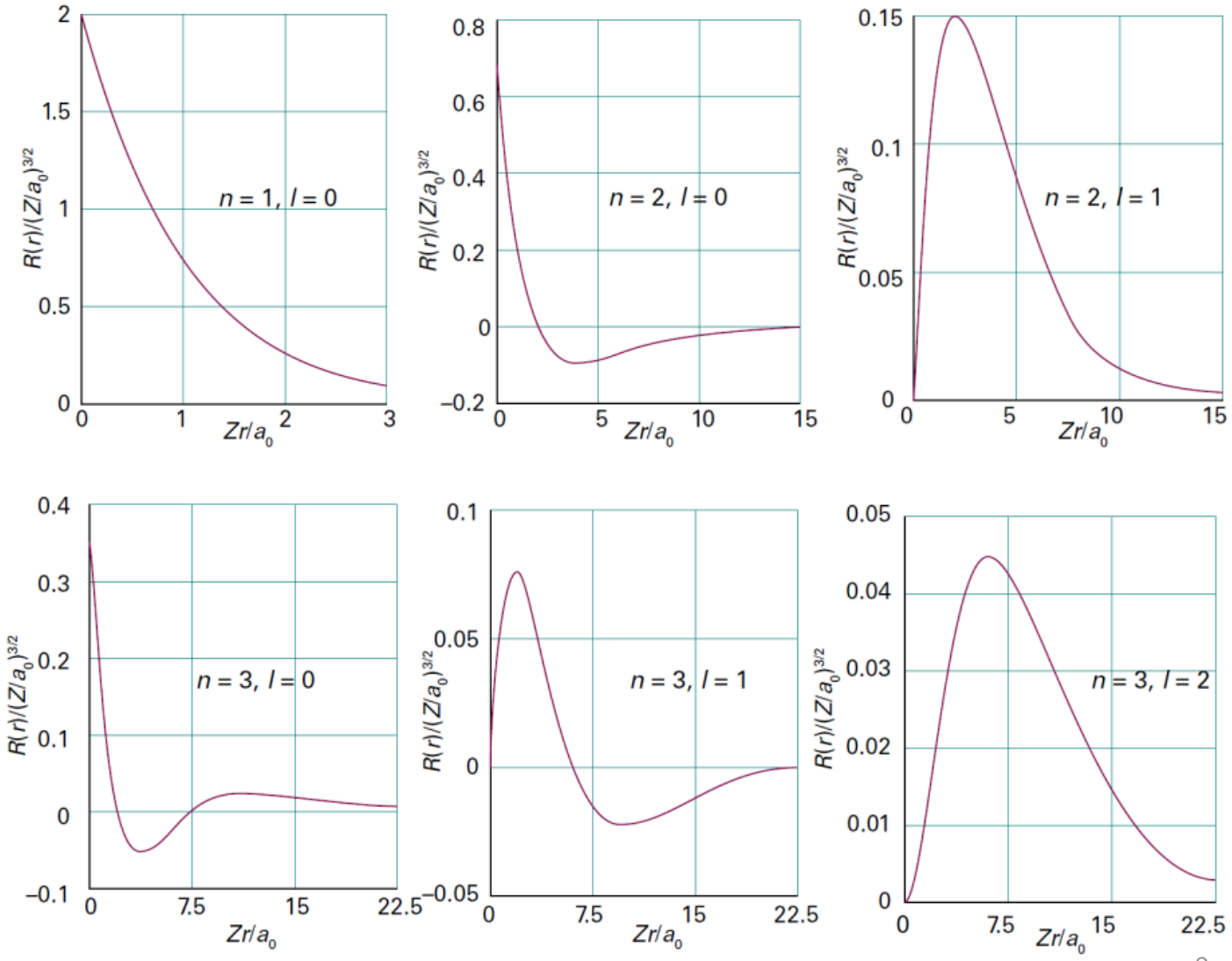
几点说明:

- 基态没有径向节点
- 激发态也可以没有径向节点（为什么？）
- 节点数 $= n - l - 1$
- $r = 0$处不算节点，因为r不能取负，$r = 0$是端点

全波函数

$$\psi_{n,l,m_l}(r,\theta,\phi) = R_{n,l}(r) Y_{l,m_l}(\theta,\phi)$$

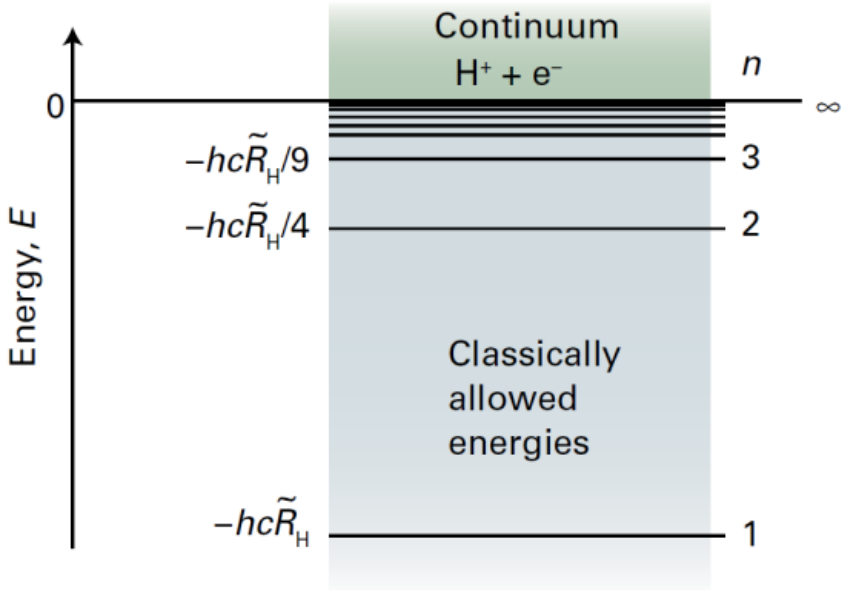
径向波函数



原子轨道：原子的单电子波函数

三个量子数：

- **主量子数 $n = 1, 2, 3, \cdots$ ：决定能量**
- **角量子数 $l = 0, 1, 2, \cdots, n\text{-}1$ ：决定轨道角动量**
- **磁量子数 $m_l = 0, \pm1, \pm2, \pm l$：决定角动量分量**



注意：以上为束缚态，量子化

$$E_n = -\frac{hcZ^2\tilde{R}_N}{n^2} \qquad \tilde{R}_N = \frac{\mu e^4}{8\varepsilon_0^2 ch^3}$$

Bound-state energies

$$\tilde{R}_N = \frac{\mu}{m_e} \times \tilde{R}_\infty \qquad \tilde{R}_\infty = \frac{m_e e^4}{8\varepsilon_0^2 h^3 c}$$

Rydberg constant

非束缚态能量>0，连续变化

里德堡常数理论值和实验值几乎吻合，偏差来源于**相对论效应**

电离能I：电离态与基态能量差　　**H原子电离能是13.6eV**

氢原子非束缚态波函数

文章引自

R. Sakuma, F. Aryasetiawan,
Phys. Rev. A. 85, 042509 (2012)

该文[15]引用了

L. D. Landau and E. M. Lifshitz,
Quantum Mechanics
(Pergamon, New York, 1977).

The continuum eigenstates with energy ϵ are labeled by continuous variable $k = \sqrt{2\epsilon}$. Their analytic expression is [15]

$$\phi_{kl}(r) = c(k,l)\frac{(2kr)^l e^{-ikr}}{(2l+1)!} F\left[\frac{i}{k} + l + 1, 2(l+1), 2ikr\right], \quad (3)$$

with $c(k,l)$ as the normalization constant,

$$c(k,l) = \sqrt{\frac{8\pi k}{1 - e^{-2\pi/k}}} \prod_{p=1}^{l} \sqrt{p^2 + \frac{1}{k^2}}, \quad (4)$$

and $F(\alpha,\gamma,z)$ is the confluent hypergeometric function. The continuum states are normalized as

$$\int_0^\infty \phi_{kl}(r)\phi_{k'l}(r) r^2 dr = 2\pi\delta(k - k'). \quad (5)$$

With these eigenstates, the completeness relation is written as

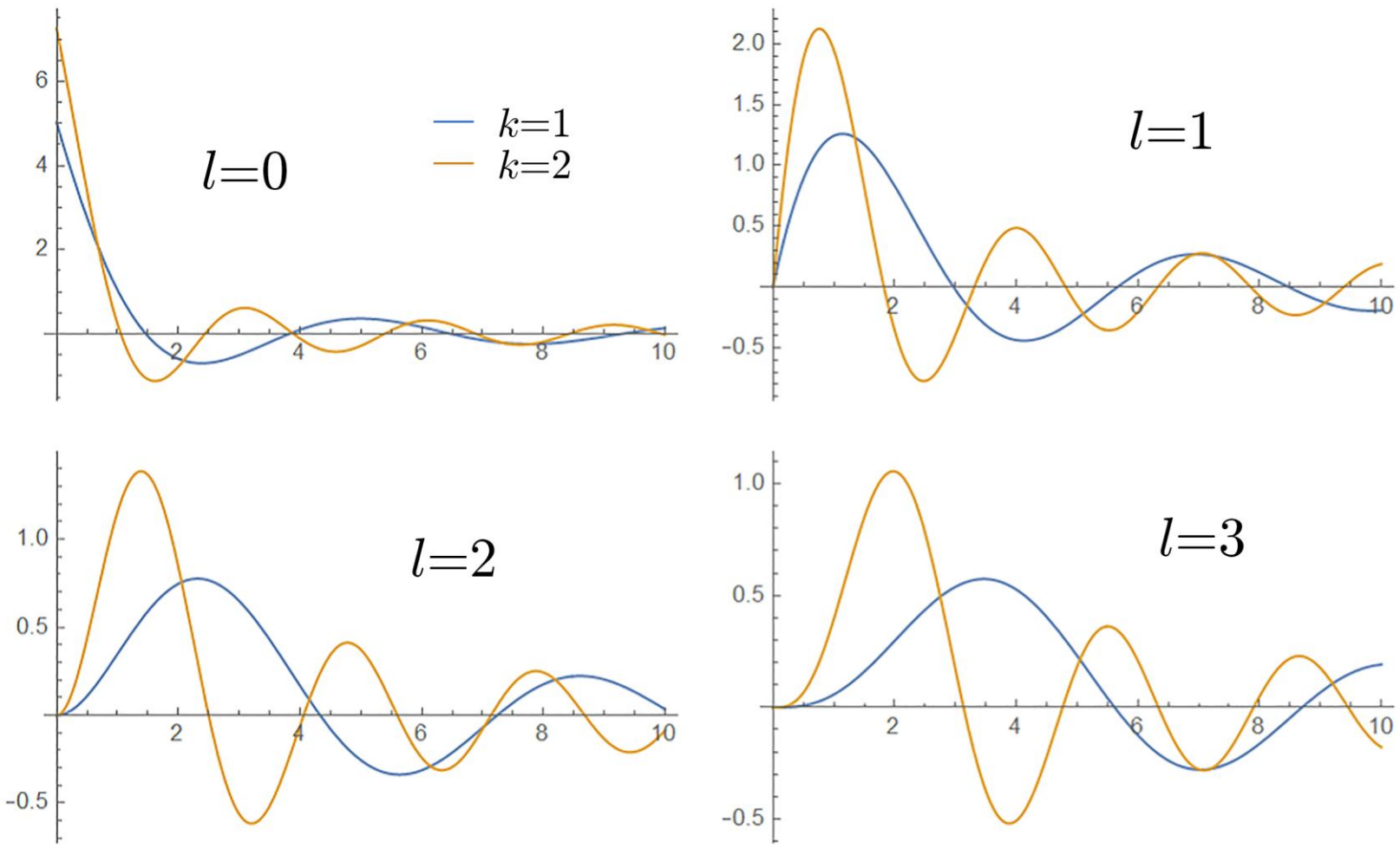
$$\delta(\mathbf{r} - \mathbf{r}') = \sum_{l=0}^{\infty}\left[\sum_n \phi_{nl}(r)\phi_{nl}(r') + \int_0^\infty \frac{dk}{2\pi}\phi_{kl}(r)\phi_{kl}(r')\right] \times \sum_{m=-l}^{+l} Y_{lm}(\hat{r})Y^*_{lm}(\hat{r}'). \quad (6)$$

非束缚态波函数$\phi_{kl}(r)$

实值函数

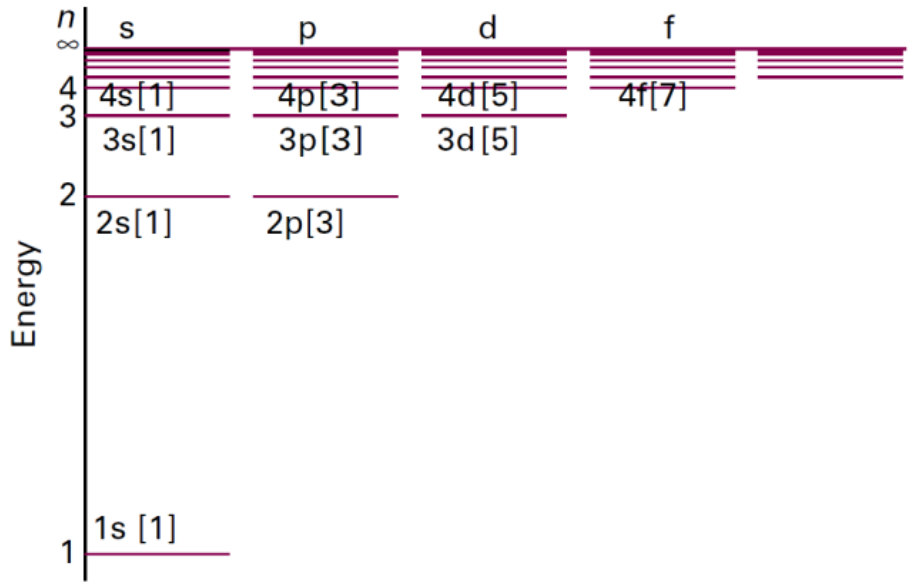
k 越大，能量越大，波长越短，振荡越快

l 增大，节点变稀疏



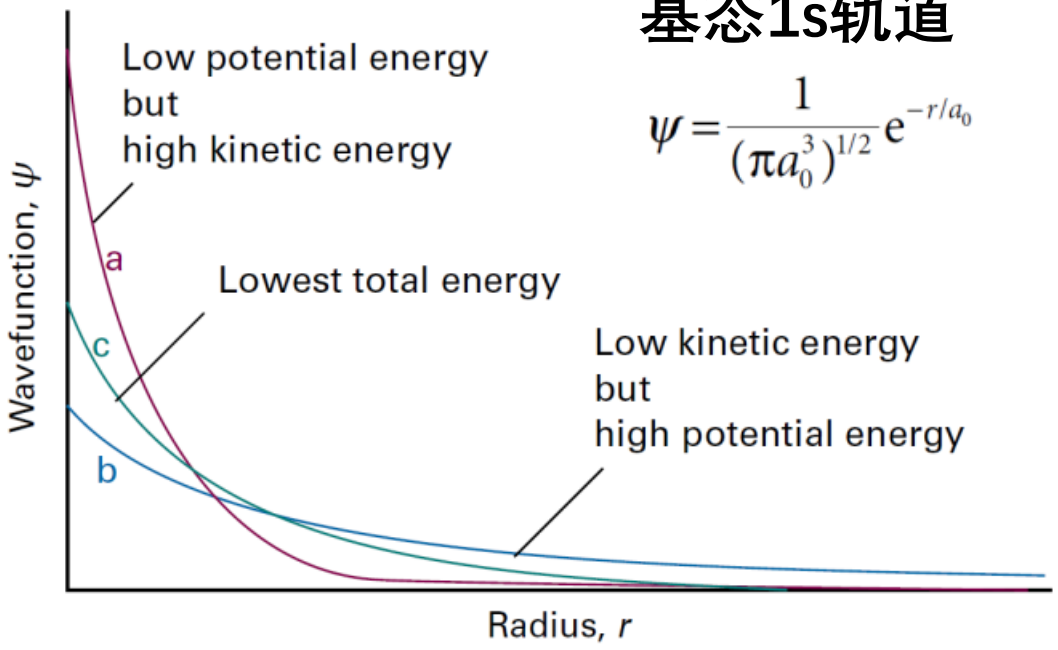
壳层(shells)与亚层(subshells)

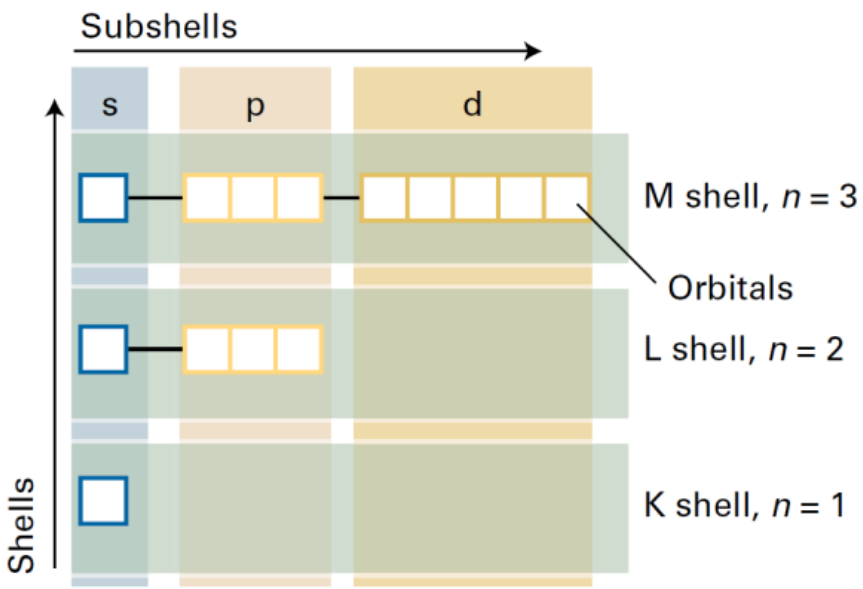
壳层	$n=$	1	2	3	4…			
		K	L	M	N…			
亚层	$l=$	0	1	2	3	4	5	6…
		s	p	d	f	g	h	i…



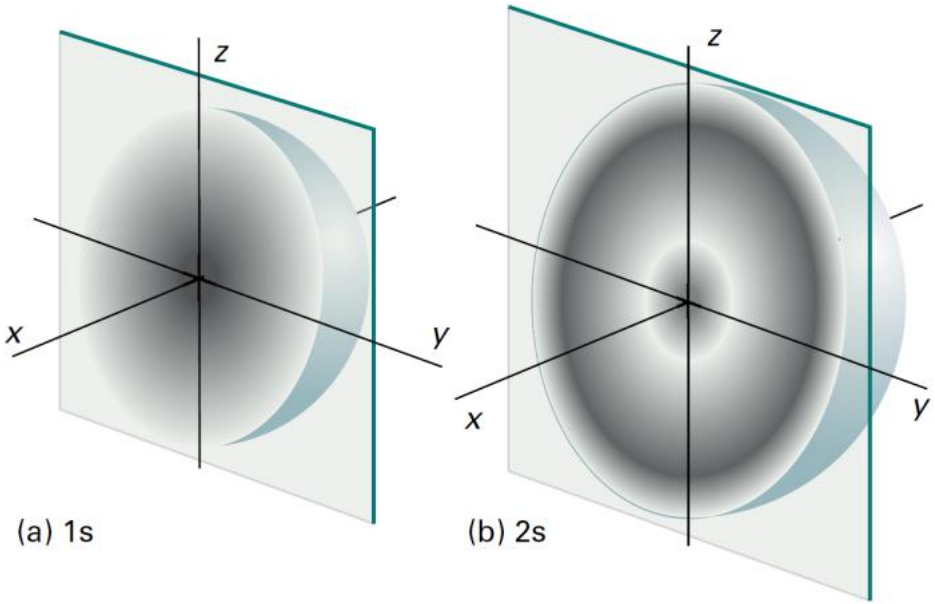
基态1s轨道

$$\psi = \frac{1}{(\pi a_0^3)^{1/2}} e^{-r/a_0}$$





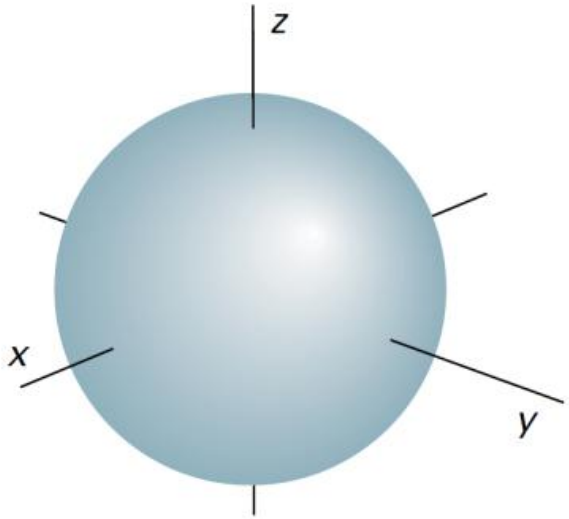
s轨道电子云(密度)截面图



轨道的平均半径：1s<2s<3s

$$\langle r \rangle = \int \psi^* r \psi \, d\tau = \int r |\psi|^2 \, d\tau$$

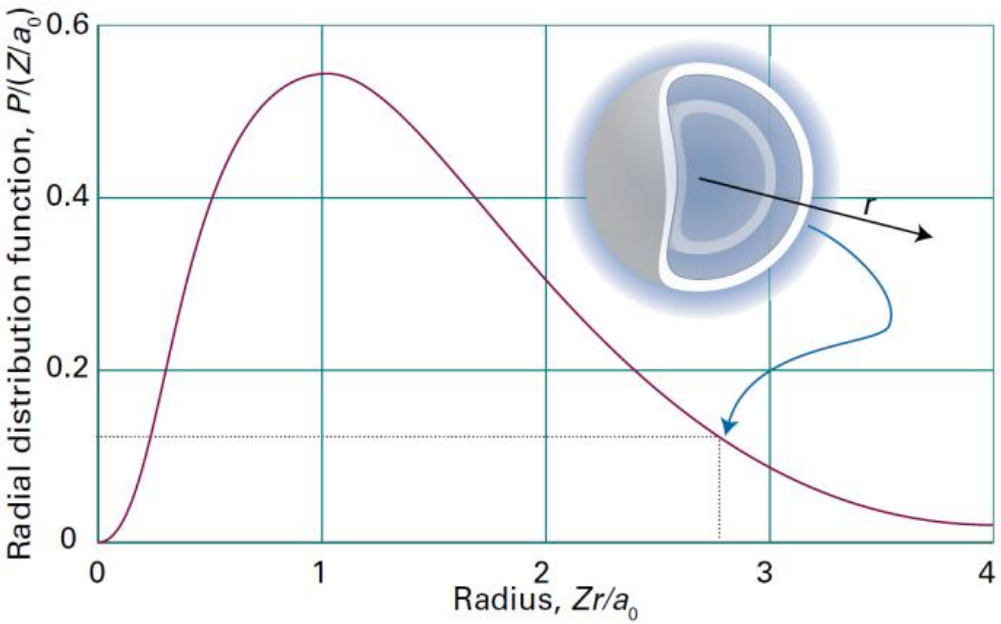
1s轨道边界面图



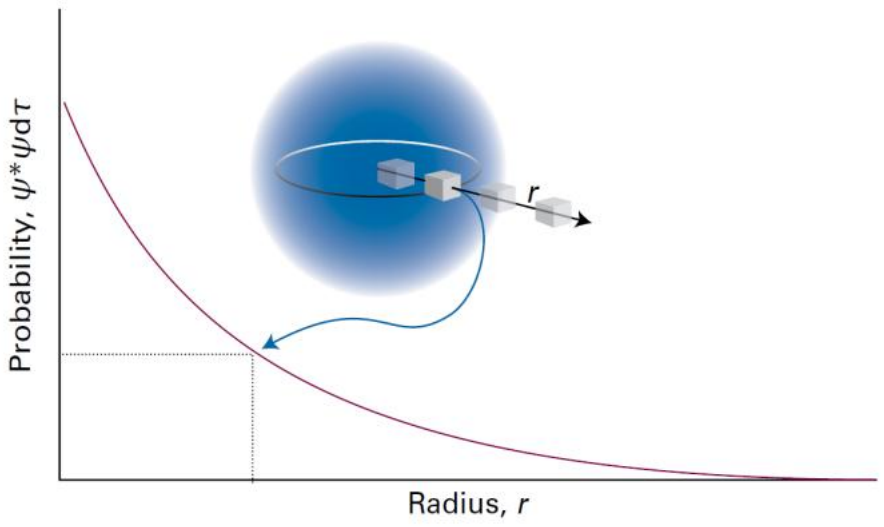
边界面(boundary surface)：保留轨道的形状，按比例放缩，包络90%的电子云

s轨道边界面图都是球面

径向分布函数 $P(r)=r^2R(r)^2$



径向概率密度



径向分布函数的性质：

- $r = 0$, $P(r) = 0$
- $r \to \infty$, $P(r) \to 0$
- $P(r)$ 极大值出现在某个中间位置，定义为轨道半径

$$\frac{dP}{dr}=\frac{4Z^3}{a_0^3}\left(2r-\frac{2Zr^2}{a_0}\right)e^{-2Zr/a_0}=\frac{8rZ^3}{a_0^3}\left(1-\frac{Zr}{a_0}\right)e^{-2Zr/a_0}$$

基态半径： $r_{mp}=\frac{a_0}{Z}$

	H	He^+	Li^{2+}	Be^{3+}	B^{4+}	C^{5+}	N^{6+}	O^{7+}	F^{8+}	Ne^{9+}
r_{mp}/pm	52.9	26.5	17.6	13.2	10.6	8.82	7.56	6.61	5.88	5.29

p轨道

$l = 1$, 有方向性，例如2p轨道

$$\psi_{2,1,0} = R_{2,1}(r)Y_{1,0}(\theta,\phi) = \frac{1}{4(2\pi)^{1/2}}\left(\frac{Z}{a_0}\right)^{5/2} r\cos\theta\, e^{-Zr/2a_0} \quad (8A.19a)$$

$$= r\cos\theta f(r)$$

$$\psi_{2,1,0} = zf(r)$$

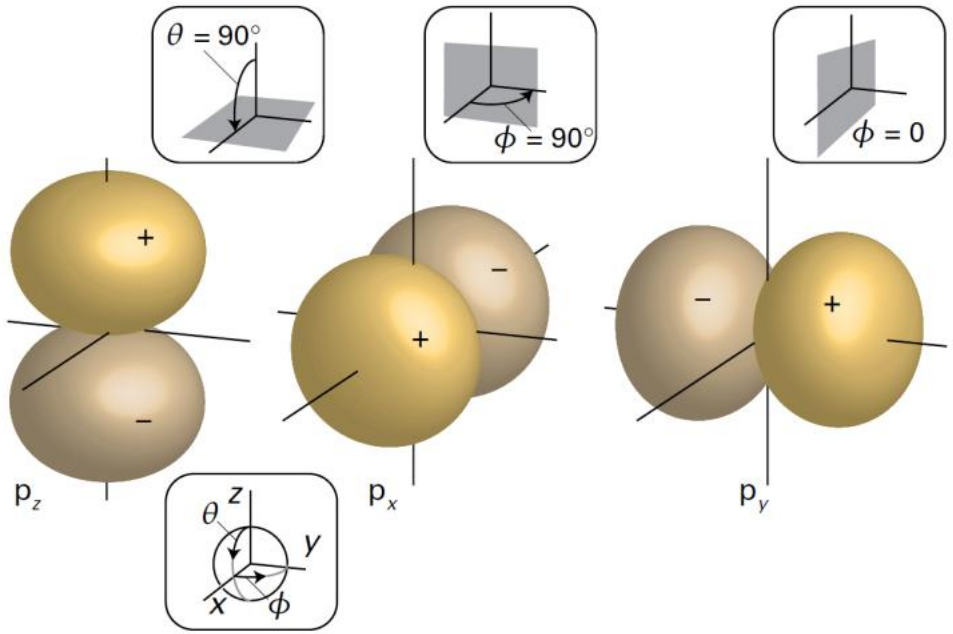
$2p_z$轨道，有节面$z = 0$ (xy平面)

$$\psi_{2,1,\pm1} = R_{2,1}(r)Y_{1,\pm1}(\theta,\phi) = \mp\frac{1}{8\pi^{1/2}}\left(\frac{Z}{a_0}\right)^{5/2} r\sin\theta e^{\pm i\phi} e^{-Zr/2a_0} \quad (8A.20)$$

$$= \mp\frac{1}{2^{1/2}} r\sin\theta e^{\pm i\phi} f(r)$$

$2p_\pm$ 轨道，是角动量 z 分量的本征态，但不是实函数

2p轨道边界面



$e^{i\phi} + e^{-i\phi} = 2\cos\phi$

$$\psi_{2p_x} = \frac{1}{2^{1/2}}(\psi_{2,1,+1} - \psi_{2,1,-1}) = r\sin\theta\cos\phi f(r) = x\, f(r)$$

$e^{i\phi} - e^{-i\phi} = 2i\sin\phi$

$$\psi_{2p_x} = \frac{i}{2^{1/2}}(\psi_{2,1,+1} + \psi_{2,1,-1}) = r\sin\theta\sin\phi f(r) = y\, f(r)$$

p_x, p_y是$p_\pm$轨道的组合(叠加)，是实函数

d轨道

l = 2, 有五种方向，例如3d轨道

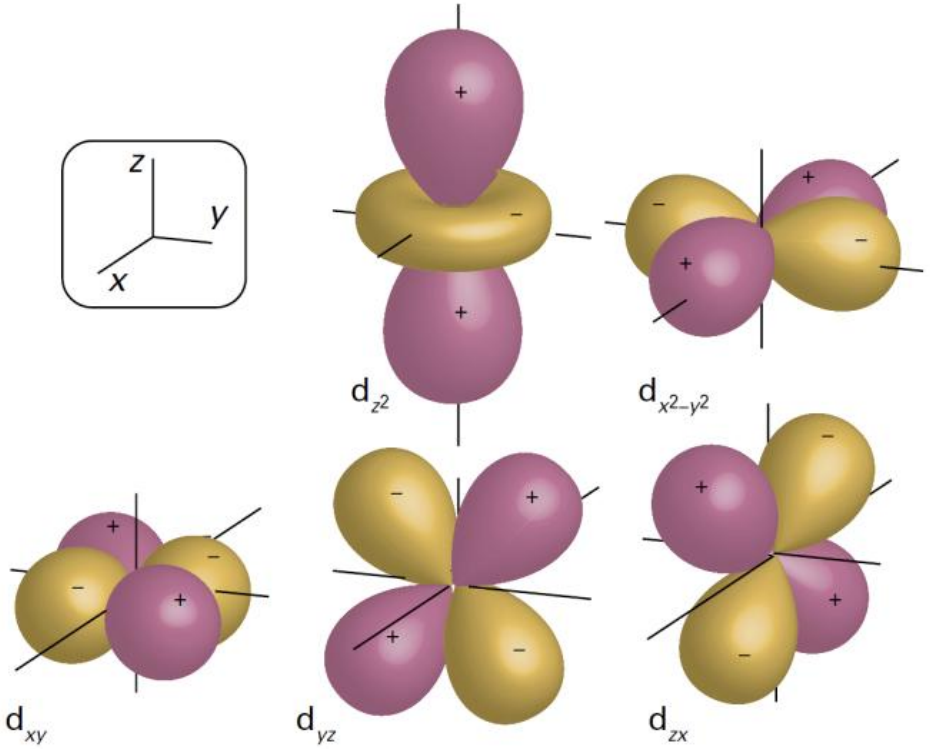
$$\psi_{d_{xy}} = xyf(r) \quad \psi_{d_{yz}} = yzf(r) \quad \psi_{d_{zx}} = zxf(r)$$

$$\psi_{d_{x^2-y^2}} = \tfrac{1}{2}(x^2-y^2)f(r) \quad \psi_{d_{z^2}} = \frac{3^{1/2}}{2}(3z^2-r^2)f(r)$$

与p轨道类似，可以组合成实函数

- d_{z^2}的形状比较另类，波函数并不正比于z^2，节面为锥面
- 其他d轨道节面为平面，轮廓成花瓣形

3d轨道边界面



习题：计算各3d轨道边界面围成的正负区域的体积

原子单位

$$-\frac{1}{2}\nabla^2\psi - \frac{Zm_e e^2}{4\pi\epsilon_0\hbar^2 r}\psi = \frac{m_e}{\hbar^2}E\psi \quad \Leftarrow \quad -\frac{\hbar^2}{2m_e}\nabla^2\psi - \frac{Ze^2}{4\pi\epsilon_0 r}\psi = E\psi$$

$$-\frac{1}{2}\nabla^2\psi - \frac{Z}{a_0 r}\psi = \frac{m_e}{\hbar^2}E\psi \quad \Leftarrow \quad a_0 = \frac{4\pi\epsilon_0\hbar^2}{m_e e^2}$$

$$-\frac{a_0^2}{2}\nabla^2\psi - \frac{Za_0}{r}\psi = \frac{m_e a_0^2}{\hbar^2}E\psi \quad \Rightarrow \quad -\frac{1}{2}\nabla_{\tilde{\boldsymbol{r}}}^2\psi - \frac{Z}{\tilde{r}}\psi = \tilde{E}\psi$$

$$(\tilde{x},\tilde{y},\tilde{z}) = \frac{1}{a_0}(x,y,z), \qquad \tilde{\boldsymbol{r}} = \frac{1}{a_0}\boldsymbol{r} \qquad \tilde{E} = \frac{1}{\frac{\hbar^2}{m_e a_0^2}}E = \frac{1}{\frac{m_e e^4}{4h^2\epsilon_0^2}}E$$

原子单位下H原子1s能量 $\tilde{E} = -\frac{1}{2}$ $\qquad a_0 = 1, \quad m_e = 1, \quad \hbar = 1$

能量单位：Hartree　　长度单位：Bohr